

T5Gemma v1 2B-2B UL2-IT vs Gemma-2-2B-IT on TReB

Same-lineage architecture comparison — 1000 samples — DeepSeek-v3.2 judge

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TL;DR

On **1000 stratified TReB English samples**, T5Gemma v1 (encoder-decoder, 5.6B) beats its same-lineage decoder-only counterpart Gemma-2-2B-IT (2.6B) on overall correctness (**35.6 % vs 30.7 %**), mean judge score (**4.47 vs 4.02**), and 16 of 26 tasks. The encoder-decoder’s advantage is concentrated in **retrieval and fact-checking** (+12 pp and +11 pp), consistent with the earlier 250-sample read but now with $4\times$ the statistical power and a reproducible, API-based judge.

Encoder-decoder costs **$2\times$ the parameters** for +5 pp CORRECT, and Gemma-2 still wins **Understanding by +15 pp**, so architecture choice is a task-mix question rather than a blanket win. Both models are wrong on **55 %** of samples — the 2–5B capacity ceiling reproduces.

Setup

Dataset	TReB English, phase-0 filtered ($\leq 4K$ tokens), stratified 1000 (seed=42)
Model A	<code>google/t5gemma-2b-2b-ul2-it</code> — enc-dec, 5.6B, 2026 (T5Gemma v1)
Model B	<code>google/gemma-2-2b-it</code> — dec-only, 2.6B, 2024 (Gemma-2 lineage)
Mode	TCoT — text chain-of-thought, <code>{"answer": x}</code> JSON
Generation	greedy, <code>max_new_tokens=256</code> , chat template, <code>eos_token_id=[1, 107]</code>
Judge	DeepSeek v3.2 via OpenRouter, 12-mode taxonomy, score 0–10
Hardware	$1\times$ H100 80GB HBM3, EU-NL-1
Wall-clock	T5G ~ 80 min @ bs=1, G2 ~ 15 min @ bs=4, judge ~ 18 min @ 12 workers

Both models share the same Gemma-2 pretraining base and the same Google-published SFT + RLHF recipe. T5Gemma v1 is described in its tech report as “an encoder-decoder adapted from Gemma-2 via UL2 + SFT,” so **architecture is the only dimension that moves** between them.

Headline

Metric	T5Gemma v1	Gemma-2-2B-IT	Δ
n	1000	1000	—
CORRECT %	35.6 %	30.7 %	+4.9 pp
CORRECT + PARTIAL %	50.0 %	44.8 %	+5.2 pp
Mean judge score (0–10)	4.47	4.02	+0.45
EM	0.117	0.092	+0.025
ROUGE-L	0.272	0.228	+0.044
Numeric-match	0.289	0.252	+0.037

Same direction as the 250-sample read, with tighter error bars. At $n = 1000$ this is past chance noise — the 2-way bucket table below makes the sign concrete.

2-way correctness buckets

Each sample classified by whether each model was judged CORRECT:

Bucket	Count	Share
TG — both CORRECT	214	21.4 %
T- — only T5Gemma CORRECT	142	14.2 %
-G — only Gemma-2 CORRECT	93	9.3 %
-- — both wrong	551	55.1 %

T5Gemma wins on net by **49 samples** ($14.2 - 9.3 = 4.9$ pp), matching the CORRECT delta in the headline. **55 % of samples are wrong for both** — the 2–5B capacity ceiling reproduces at $4\times$ the sample size.

Failure-mode distribution

Mode	T5Gemma v1	Gemma-2-2B	
CORRECT	35.6 %	30.7 %	T5G +4.9 pp
WRONG_SELECTION	35.8 %	39.0 %	dominant failure for both
PARTIAL	14.4 %	14.1 %	tie
NUMERIC	3.9 %	3.0 %	
HALLUCINATION	3.4 %	3.8 %	
TRUNCATED	1.9 %	2.2 %	much lower than 250-run’s 17.7 %
REASONING	1.7 %	1.1 %	
OTHER	1.2 %	0.7 %	
REFUSAL	1.0 %	3.7 %	G2 refuses $\sim 4\times$ more
FORMAT	0.7 %	0.7 %	tie
MISUNDERSTAND	0.3 %	0.3 %	tie
EMPTY	0.1 %	0.7 %	G2 has 7 silent no-ops

Two modes track architecture:

- **Gemma-2 refuses $\sim 4\times$ more** (3.7 vs 1.0 %) — same signal as the 250-run. Same SFT pipeline, so almost certainly a decoder-only vs encoder-decoder calibration difference.
- **T5Gemma TRUNCATED fell to 1.9 %** (from 17.7 % at 250). The 250-set over-indexed on long-output Table_Summary / retrieval; stratified 1000 doesn't. T5Gemma is *not* a structurally long generator — max_new_tokens=256 is enough.

Per-category performance

Tasks grouped into 7 semantic categories:

Category	n	T5G %	G2 %	T only	G only	both OK	both wrong	Winner
retrieval	192	41.1	29.2	39	16	40	97	T5G
fact_checking	204	52.9	42.2	48	26	60	70	T5G
computation	223	32.3	28.7	26	18	46	133	T5G
understanding	84	45.2	60.7	6	19	32	27	G2
summarization	158	32.9	27.8	18	10	34	96	T5G
analysis	104	2.9	2.9	3	3	0	98	tie (floor)
multi_step	35	11.4	8.6	2	1	2	30	T5G (small n)

Reads:

- **Retrieval + fact-checking** ($n = 396$, $\sim 40\%$ of the benchmark) is T5Gemma's strongest advantage. Mechanistically consistent with cross-attention — the decoder re-attends to the encoded table at every step, which is the shape of “find this specific thing” and “does this claim entail from the table.”
- **Understanding** ($n = 84$) — **Gemma-2 wins +15 pp.** These tasks reward generic instruction-following and concise rephrasing. Mostly reference-free (judged by constraint satisfaction) where decoder-only SFT calibration does better.
- **Analysis** ($n = 104$) — **both at 2.9 %.** Distribution testing, hypothesis testing, outlier detection, correlation analysis all need numeric reasoning neither 2–5B model has. Reinforces the 2–5B capacity ceiling.

Per-task deltas ($|\Delta|$ mean score ≥ 1.0)

Task	n	T5G	G2	Δ	Winner
Understanding	64	5.16	7.97	-2.81	G2
Hallucination_Evaluation	63	5.67	3.73	+1.94	T5G
Instruction_Following	20	2.50	0.67	+1.83	T5G
Table_Outlier_Detection	20	3.58	1.95	+1.63	T5G
Table_Selection	64	3.73	2.18	+1.55	T5G
Table_Plausibility_Verification	15	4.29	2.86	+1.43	T5G
Table_Query	64	3.65	2.26	+1.39	T5G
Mathematical_Reasoning	63	4.95	6.23	-1.27	G2
Table_Domain-specific_Operations	34	3.53	2.34	+1.19	T5G
Table_Fact_Checking	63	7.58	6.48	+1.11	T5G
Table_Hypothesis_Testing	20	1.58	2.65	-1.07	G2

Scoreboard (mean score, all 26 tasks): T5G wins 16, G2 wins 6, near-ties ($|\Delta| < 0.3$) = 4. Gemma-2’s wins cluster in **general-reasoning** tasks (Understanding, Mathematical_Reasoning, Hypothesis_Testing) where a well-calibrated CoT helps more than cross-attention.

Take-aways

1. **Encoder-decoder helps retrieval + fact-checking at 2–5B scale.** +12 pp retrieval, +11 pp fact-checking. Cross-attention is a real architectural lever for table lookup.
2. **It’s not free — 2× params for +5 pp overall.** If your task mix is heavily lookup / entailment / cell-selection, T5Gemma pays for itself. For open-ended understanding or math CoT, pick Gemma-2.
3. **2–5B ceiling is load-bearing.** 55 % both-wrong; analysis floored at $\sim 3\%$; multi-step at $\sim 10\%$. Either scale past 4B or design expectations around this.
4. **Gemma-2 REFUSAL quirk is not noise.** 3.7 % vs 1.0 % reproduces at larger n . Same Google SFT pipeline, so the gap is decoder-only calibration — or UL2 re-cal — specific.
5. **The 250-run TRUNCATED (17.7 %) was a sampling artifact.** Stratified 1000 gives 1.9 %. No reason to raise max_new_tokens above 256 for T5Gemma v1.

Follow-ups (deferred)

- **Gemma-2-9B-IT** as a param-equal comparator (5.6B enc-dec vs ~ 9 B dec-only) — tells us whether the retrieval advantage is architecture or capacity.
- Probe the SWA-bug threshold on T5Gemma v1; if v1 is bug-free we can relax the 4K-token filter and recover the Multi-step tail.

- Pull the 37 Gemma-2 refusals and look for a systematic pattern (task-type, table length, specific prompt shape).

Appendix — representative samples

Two per bucket (TG / T- / -G / --), seed=42 random pick over 1000 pairs. Predictions truncated to last 250 chars.

T5G only (T-)

#1 — **Table_Selection** — T5Gemma CORRECT (10), Gemma-2 WRONG_SELECTION (0).

Q: What is “synthesis” where “synthesis” > 3.016 and “inflection” > 0.33? **Gold:** 3.7.

T5G: Correctly identifies Eskimo (synthesis=3.70, inflection=0.34); emits {"answer": 3.70}.

G2: Walks rows ruling each one out, never finds Eskimo, no {"answer"} emitted.

#2 — **Hallucination_Evaluation** — T5Gemma CORRECT (10), Gemma-2 WRONG_SELECTION (0).

Q: Pick the more realistic continuation for a news lead about Xiaohai Lake’s ecology. **Gold:** (A).

T5G: (A) — “directly relates to observed bird activity and lush vegetation.”

G2: (B) — overweights the “jewel of the high plateau” framing.

Gemma-2 only (-G)

#3 — **Table_Fact_Checking** — T5Gemma WRONG_SELECTION (0), Gemma-2 CORRECT (10).

Q: “DFDS operates in Denmark only” — Entailment / Neutral / Contradiction. **Gold:** Contradiction.

T5G: Reads “Baltic Sea, English Channel, North Sea,” then says *Entailment*. Logic-flip.

G2: Same rows → Contradiction.

#4 — **Table_Retrieval** — T5Gemma OTHER (judge parse-fail), Gemma-2 CORRECT (10).

Q: Who has the average 50.16? **Gold:** Brad Hodge.

T5G: {"answer": "Brad Hodge"} — semantically right, judge JSON didn’t parse cleanly.

G2: Same answer, parsed cleanly → CORRECT.

Both correct (TG)

#5 — **Table_General_Operations** — both CORRECT.

Q: Difference in points between highest and lowest Higher-level grades. **Gold:** 100.

Both walk H1 (100) and H8 (0), compute 100, emit {"answer": 100}.

#6 — **Table_Fact_Checking** — both CORRECT.

Q: “Airbnb was founded on August 2001.” **Gold:** Contradiction (table says 2008).

Both spot the year mismatch, both emit Contradiction.

Both wrong (--)

#7 — **Table_Plausibility_Verification** — T5G PARTIAL (5), G2 WRONG_SELECTION (0).

Q: Identify population anomalies. **Gold:** Farrer (too high) and O'Malley (too low).

T5G: O'Malley + Phillip (wrong second row) — half credit.

G2: O'Malley via demographics; misses Farrer.

#8 — **Table_Distribution_Testing** — T5G WRONG_SELECTION (0), G2 TRUNCATED (0).

Q: Does the “Income (million euros)” column follow normal? **Gold:** Yes.

T5G: Hedges “insufficient information”; emits `{"answer": "Insufficient information provided."}`.

G2: Enumerates Q-Q plots, Shapiro-Wilk, etc. then truncates before answering.

Reproducibility: `sample_1000.py -seed 42` → `dataset/sample_1000.json`. Predictions in `results/{t5gemma_v1_`
Merged judged output in `results/judged_1k_2way.jsonl` (2000 rows). Judge: `score.py -judge`
`deepseek-v3.2`.